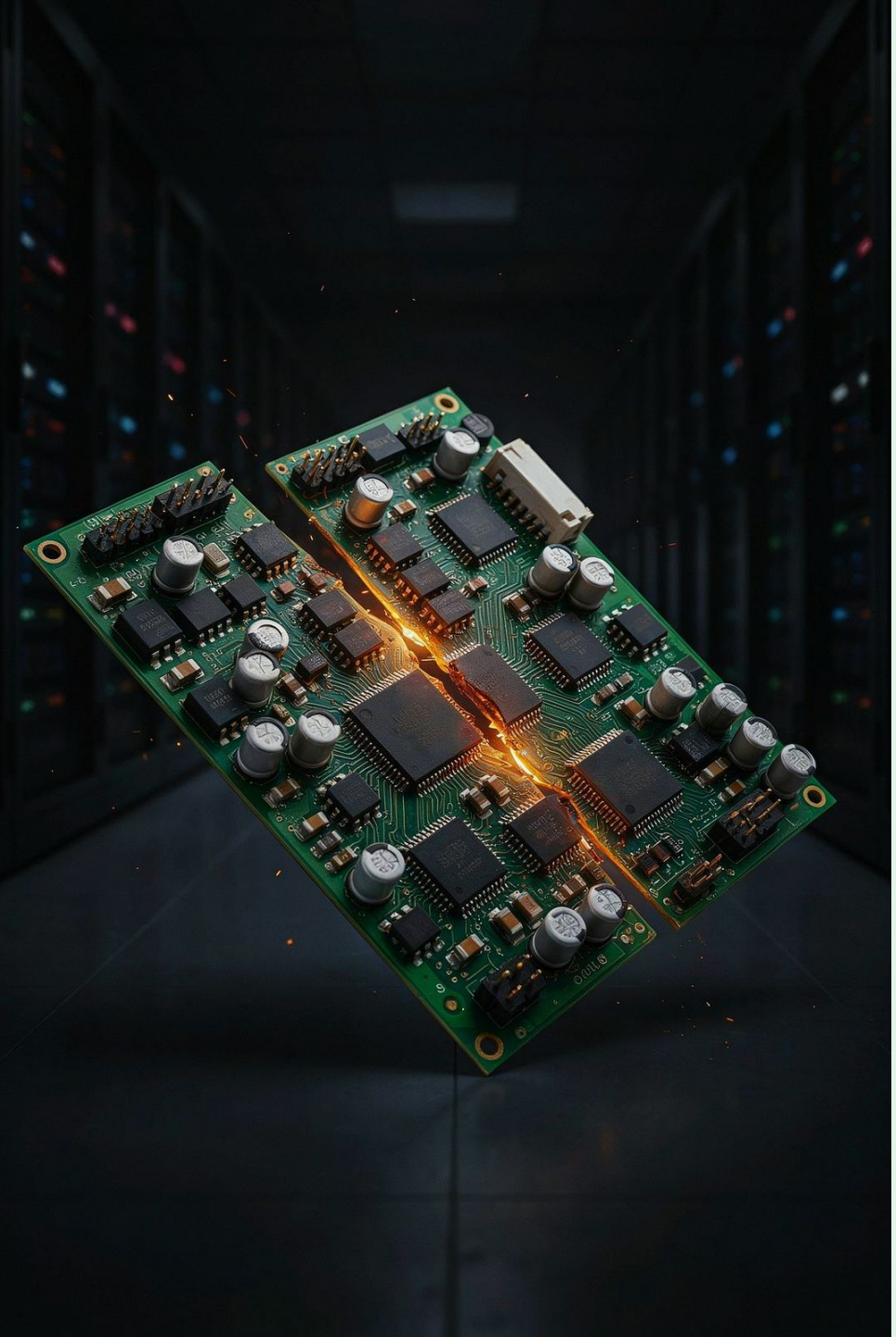


Why AI Systems Fail

MESA Term 2 - Cohort C4

Lecture 1 - October 21, 2026

The Shift

In Term 1, you built four AI projects - prompts, agents, automations. Those builds are now your case studies.

Term 2 question: how do you decide **what** to automate before you figure out **how** to automate it?

This session is about what it costs when those decisions are made wrong.



Four types of AI systems. Four different decisions.

Prompt / Generative AI

Fast to build, high output variance, hard to audit at scale.

Workflow Automation

Predictable, brittle on edge cases, high maintenance load.

AI Agent

Autonomous, high upside, largest failure surface.

Machine Learning Model

Custom-trained, accurate at scale, requires data and time.

Picking the wrong one isn't a technical mistake. It's a commissioning mistake.

What You Built in Term 1

AI Agents & Automations

You built with Claude Code - writing prompts, chaining tools, and debugging unexpected outputs.

Everything Ran

Outputs were generated. Actions were triggered. Responses came back. The builds worked.

Term 2 Starts Here

That was Term 1. Term 2 begins with a harder question.



Did your build deliver the outcome?

Not: did the AI respond? Not: did the code run?

Did the thing you built change what needed to change in the world?

⚠ At companies investing millions, the honest answer, most of the time, is **No**.

r/recruitinghell
u/Fast-Application5391 • 1h

applied to my own company's job posting under a fake name. rejected in 6 minutes.

I lead a team of nine. We've had a desperate need for a senior associate role since april and hr keeps telling me the pipeline was thin, nothing passing screening. So last month i made a new email, changed my name, tweaked the dates, and applied to my own req with essentially my own cv. The CV of the person who wrote the job description and brother i was auto-rejected in six minutes!

Took a week to work out why. the filter was set to require React.js and my cv says React. that's it. that's the whole thing. a full stop.

We have been rejecting every qualified applicant since april over punctuation, while i sat in weekly meetings being told the market was tough.

if you've been getting instant rejections from places you're clearly qualified for, it is very possibly not you.

Anyone else tested their own funnel? i'm now scared to look at the other two requirements

The Gap Nobody Talks About

The failure is not in the model. The failure is not in the code.

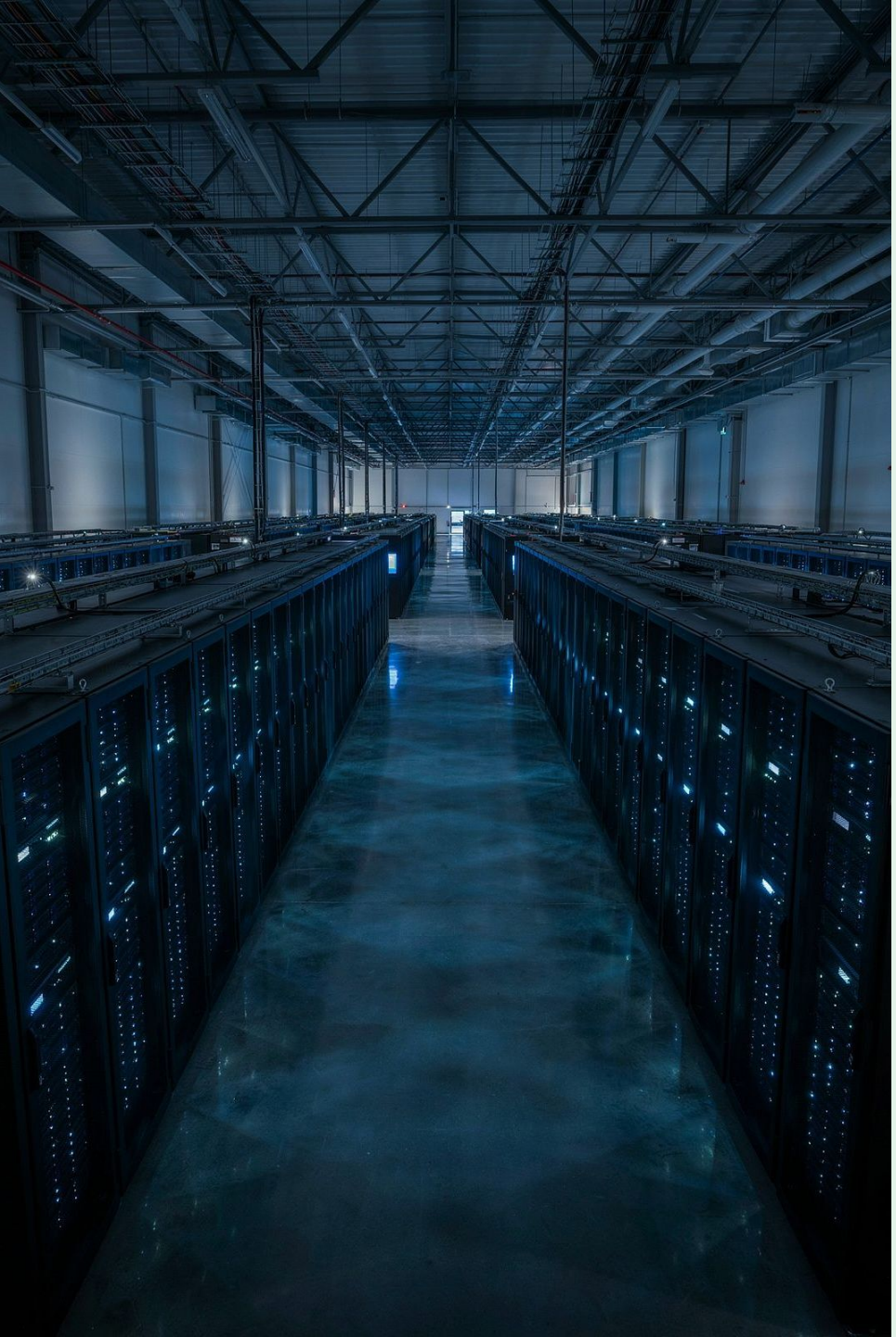
A System That Works Technically

Code runs. API calls return. The model responds. Every log shows green.

A System That Delivers the Outcome

Something measurable changes in the world. A problem is solved. A decision improves. A cost falls.

This gap is what we are designing for in Term 2.



SECTION 1

The Data

How widespread is this problem, and what does it actually look like in practice?

AI Is Everywhere. Outcomes Are Not.

72%

Have Deployed AI

Organisations with AI in at least one function - McKinsey Global Survey 2024

22%

Report Real Impact

Organisations reporting a "significant" EBIT improvement attributable to AI

<20%

Reach Production

Enterprise AI projects reaching production with measurable business outcomes - Gartner 2024

The Most Expensive Mistake

Deployment is not impact. Organisations that measure model activity - queries handled, tokens processed, requests per second - are measuring the wrong thing entirely.

The gap between deployment and demonstrated value is where most enterprise AI investment is currently lost.



CASE STUDY

Klarna's AI Customer Service Launch - February 2024

2/3

Of all customer service chats handled by AI within 30 days

700

Full-time agent equivalents replaced in a single month

11 → 2 min

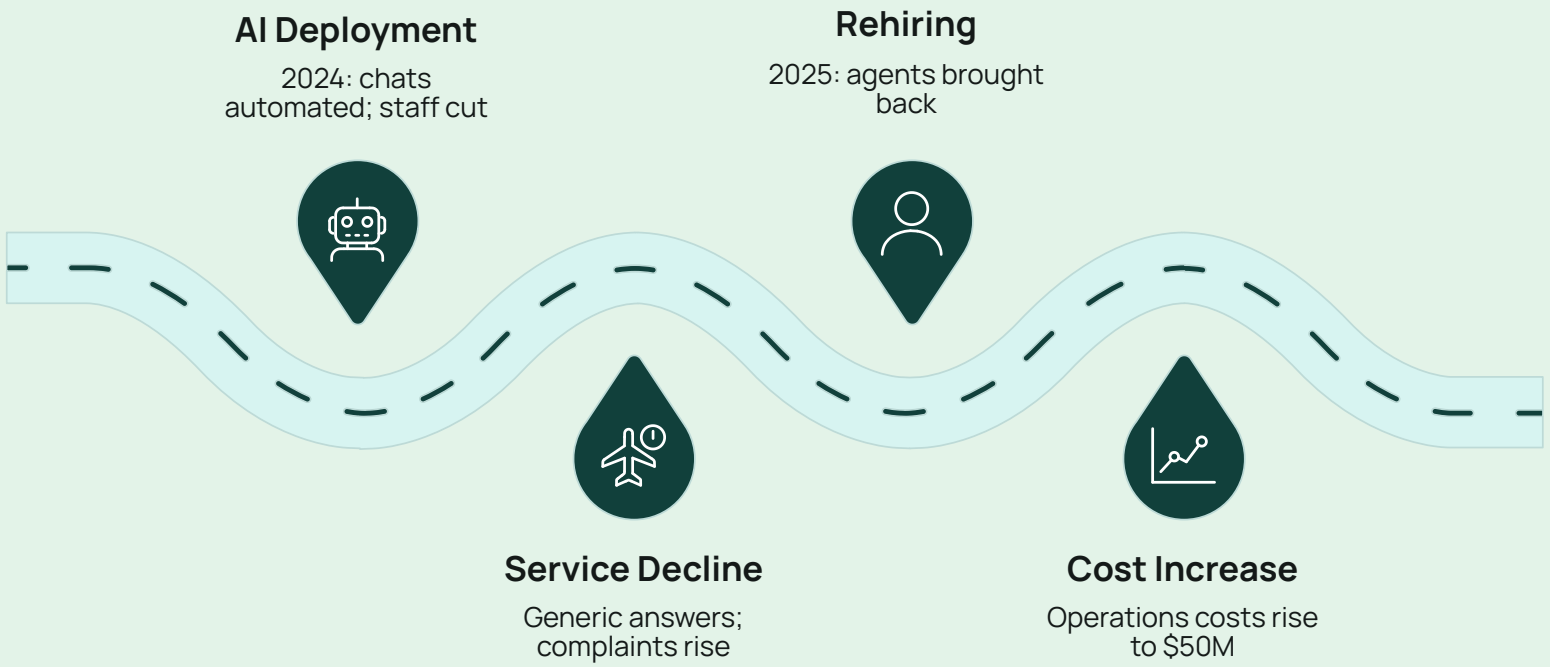
Resolution time dropped from 11 minutes to under 2

$40M

Projected profit impact announced at Davos for 2024

The tech press called it the AI success story of the year. The numbers looked decisive. The story was not over.

What the Numbers Actually Showed



The Reversal

Experienced service staff were laid off. The AI gave generic answers on complex queries. Complaints rose. By May 2025, Klarna was actively re-hiring human agents.

Customer service and operations costs rose from **$42M to $50M** in Q3 2025 - despite the claimed $60M in savings.

> "Klarna is a poster child for bad AI deployment - they overpivoted toward cost containment without weighing long-term customer experience." — *Kate Leggett, Forrester*

What Klarna Measured vs. What Mattered

What Was Measured

- Chats handled per hour
- Time to first response
- Volume equivalent to N agents

What Actually Mattered

- Problems resolved per customer
- Customers retained after interaction
- Cost per resolved issue

The Default Failure Mode

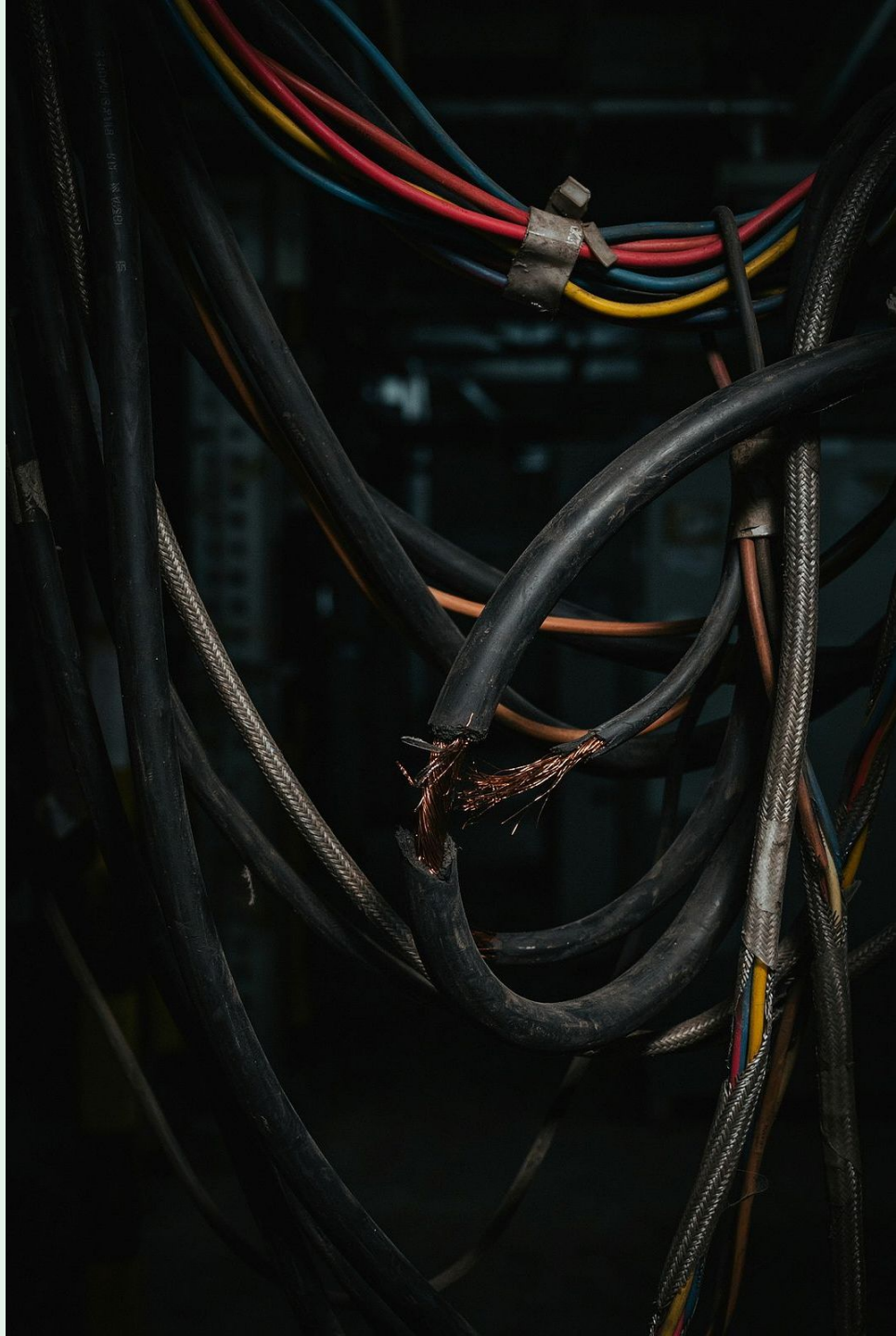
The metrics Klarna chose were **behavioural** - describing what the system did. The outcome metric - whether customers left with their problem solved - was never demonstrated.

This is not a Klarna problem. It is the default failure mode across enterprise AI deployment.

SECTION 2

Anatomy of a Failure

The system worked. The outcome didn't arrive. Why?





CASE STUDY

McDonald's + IBM Drive-Through: A 2.5-Year Pilot

1

Oct 2021

IBM AI voice ordering deployed at ~100 drive-through locations

2

2022–2023

Accuracy stalls in the low 80% range - target for wider rollout was 95%+. TikTok videos of 9 butters and hundreds of McNuggets go viral.

3

Jun 2024

McDonald's ends the IBM partnership. The pilot is terminated after two and a half years.

What Went Wrong

What the AI Did

Transcribed customer speech → generated an order entry in real time. **This worked.**

What Didn't Change

The drive-through workflow. In a human-staffed lane, the cashier reviews the order on screen before confirming - catching a misheard item before it reaches the kitchen.

The AI's output fed **directly into the kitchen queue**. No review step was added. The old workflow assumed human accuracy. The new system ran at 80%.





SECTION 3

Behavioural vs. Outcome Requirements

The brief specified what the system should *do* - not what must *change* in the world.

The Air Canada Case

Moffatt v. Air Canada - BC Civil Resolution Tribunal, February 14, 2024

1 The Query

Jake Moffatt's grandmother died. He asked Air Canada's chatbot about bereavement fares before booking his flight.

2 The Chatbot's Answer

A reduced bereavement fare is available - apply within 90 days of ticket issuance, after travel.

3 The Actual Policy

Bereavement fares do not apply to requests made after travel is completed. Air Canada refused Moffatt's claim.

4 The Defence - and the Ruling

"The chatbot is a separate legal entity responsible for its own statements." The Tribunal rejected this. Air Canada paid CAD $650 + fees + press coverage.

Two Ways to Write the Same Requirement

The Requirement That Was Written ✓

"The AI chatbot answers customer queries about Air Canada policies."

It answered. Requirement met.

The Requirement That Should Have Been Written ✗

"Customers receive information they can rely on to make financial decisions - information Air Canada will honour."

Failed. Tribunal. CAD $650 + fees + press coverage.

The Root Cause

The system did exactly what it was specified to do. The specification described a behaviour - generate a response - not an outcome: give customers accurate, actionable, binding information.

The specification was wrong. And a wrong specification, faithfully executed, is still a failure.

Behavioural vs. Outcome Requirements

Behavioural Requirement

Describes what the system **does.**

"The AI processes customer queries and returns a response."

Evaluated by looking at the system's output.

Outcome Requirement

Describes what must be **true in the world** after the system runs.

"Customers leave the interaction with accurate information they can act on."

Evaluated by looking at what changed in the world.

A system can satisfy every behavioural requirement and still fail every outcome requirement. **This is the structural cause of most AI system failures.**

How to Write an Outcome Requirement

The Test

Can you evaluate success **without looking at the AI?**

- If evaluation requires inspecting the AI's output → you have a **behavioural** requirement
- If evaluation looks at what changed in the world → you have an **outcome** requirement

The Template

> "After this system runs, **[person / process / metric]** will **[measurably change in this specific way]**."

Apply this template before writing a single prompt. If you cannot complete the sentence with specifics, you are not ready to build.

The Five Pre-Build Decisions

Before writing a single prompt, every system in Term 2 requires five decisions:

1

Outcome Definition

What must measurably change in the world?

2

Baseline

What does the current state look like, expressed in numbers?

3

Workflow Change

What in the existing process must be redesigned around the AI's actual capability ceiling?

4

Measurement Plan

How will we know - in the world, not in the logs - whether the outcome was achieved?

5

Kill Criterion

At what specific point do we stop and redesign rather than persist?

Arjun's Hiring Pipeline

1 Outcome Definition

✗ "AI scores 240 CVs accurately."

✓ "More shortlisted candidates pass Round 1 vs. the manual baseline."

2 Baseline

No past-hire data means no ground truth for the rubric. Need: outcomes of the last 8–10 hires

3 Workflow Change

Arjun moves from judge to auditor - risks automation bias. Design for it: review the AI's top 20 and spot-check 10% of rejections every session.

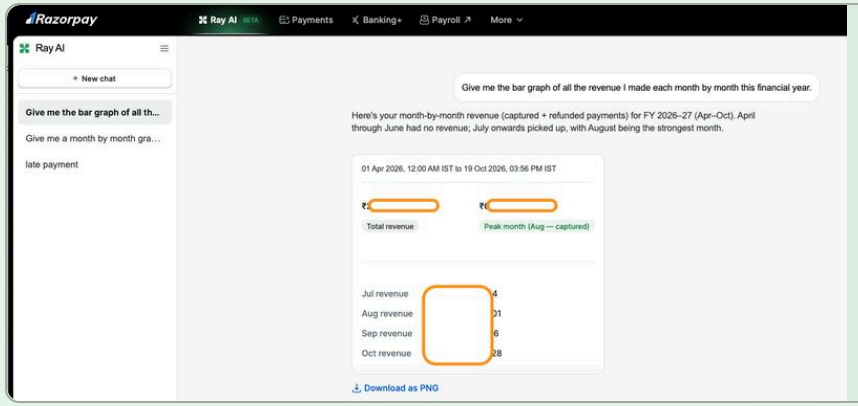
4 Measurement Plan

Primary: shortlist-to-Round-1 pass rate. Secondary: false negative rate on a sample of every batch. Instrument: "Was this candidate worth 45 minutes?"

5 Kill Criterion

If 3+ of the first 10 shortlisted are out within 10 minutes of Round 1, stop and recalibrate.

The Difference between Good & Great



LinkedIn

How NOT to implement AI ft. Razorpay Logged in to my Razorpay dash...

How NOT to implement AI ft. Razorpay Logged in to my Razorpay dashboard yesterday. The landing page is now Ray AI. A chatbot. The numbers I open this...

Write the outcome before you write the prompt.

If you cannot state what must change in the world - you are not ready to build.

The One Rule for Term 2.

